

Pest Detection in Crops Using Deep Neural Networks

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Abstract— Pests play a significant impact in crop destruction. Currently, crop yields are being reduced as a result of pest- infested crops, resulting in a reduction in output rate. The possibility of identifying the plant disease in an unfavorable environment is not pursued further. The key problem is to reduce pesticide use in the agricultural field while increasing the assembly rate's standard and quantity. This paper is laboring to explore the plant disease prediction at an untimely action. In spite of traditional convolution neural network (CNN)-based approaches, new method for detecting and recognizing insect pests have been created to produced adequate results. To optimize various parameters, CNN-based approaches necessitate a huge dataset. As a result, CNN-based two-stage recognition and identification process for insect pests is proposed. A region suggestion network for insect pest detection using YOLOv3 and a re-identification method are also presented in this paper. To train these models, knowledge augmentation method using image processing is proposed. YOLOv3 provides 92.11% accuracy when compared to CNN during pest detection.

Keywords— CNN; YOLOv3; Pest Detection; Crop; Yield

I. INTRODUCTION

Agriculture plays a crucial part in the economy and is regarded as the backbone of the economic system in emerging countries. Agriculture is critical to human survival because it ensures food security [1]. Pests are thought to be responsible for up to one third of all agricultural output. Pests have a significant impact on worldwide agricultural economy since they are known to cause a variety of crop damage. Insect pest identification is crucial for agricultural pest forecasting, which is necessary for healthy agricultural economics and food security. Identifying insect pests is critical in agricultural pest forecasting, which is vital for stable agricultural frugalities. Pests' detection is a very important role to check the infected leaf in the agriculture field using some prediction technique.

Crop health is also monitored by drones equipped with cameras. Using those drone images, the pests affected leaves are identified and the proper usage of pesticides improves the crop yield [2]. Periodically, images of pest-affected leaves are taken with a drone equipped with a camera. The pest images on the leaves are identified using CNN and

YOLOv3 [3]. In the proposed method, CNN is used to determine whether the

obtained plant photos contain diseases or pests. The drones captured images with sufficient resolution is used to make accurate identification automatically and generating an audio to giving a details of health and bacterial leaf spot/disease on the plants [4] [5].

The proposed system consists of the following steps,

- Drone images must be acquired at sufficient resolutions (of the order of one millimetre) to recognizes the pest in the plant.
- The deep CNN-based image processing system is powerful enough to analysis the problem and sufficiently well-trained to "recognize" the pests in the crop.
- For the detection task, image processing based on YOLOv3 is used.

The remainder of the paper is laid out as follows: Section II confers the related researches in the literature. In Section III, the proposed approach is presented. Section IV explains the findings and discussions.

II. RELATED WORK

This section presents some of the literature works relevant to the proposed pest detection methodology. Kulbacki et al. have presented a detailed study on the usage of drone technology for agricultural purpose [6]. Drones are used for improving Precision Agriculture (PA) effectively. The images from drone plays a vital role for pest detection and controlling in crops. Deep convolutional neural networks (DCNN) is used in [7] for finding disease in banana leaves. ResNet50 is used to find the banana disease and provides 90 % accuracy when compared to the other models in the literature such as MobileNetV1.

Abraham Chandy [8] have proposed a deep learning-based AI technique for finding pests in coconut trees. The coconut trees images are captured using drone which is having inbuilt camera and NVIDIA Tegra. 4 different types of plant diseases and 5 different types of pests were identified using this system. Fernando et al. have discussed

a drone-based technique for detecting arthropod pest in farm field [9]. The

drones have been developed from 1998, the agricultural usage of it was increased from last decade. According to some study, the drone market will be \$36.9 billion by 2022.

Gao et al. have introduced one framework for crop pest detection using IoT [10]. Pest detection is done with the help of drones which are aircraft that are not piloted (UAVs). Sensors and cameras are used in UAV and the relationship between weather pests/diseases were investigated. The proposed framework is used for both environmental monitoring and pest monitoring. Van Hulle et al. have presented deep learning based leaf image classification for pest detection [11]. The proposed method can be used to identify a variety of plant diseases and also be differentiated between plant and the surroundings. Caffe is used for learning and provides up to 98 % precision.

Yan Guo et al. have introduced deep learning-based plant disease detection [12]. Leaf localization is done with the help of the region proposal network (RPN) and the region is segmented using Chan –Vese (CV) algorithm. Further training is done with the help of the transfer learning and the accuracy of 83.57% is received. Hassan et al. have proposed CNN and transfer learning-based plant leaf disease detection methodology [13]. EfficientNetB0 provides better accuracy than the other CNN networks in the literature. But the MobileNetV2 is usable for handheld devices while having optimized factors.

III. PROPOSED WORK

Pest identification in agricultural fields has garnered a lot of attention, which is vital for smart agriculture. Automatically monitoring crop pests, in particular, is a critical component in agricultural resource management and optimization. In spite of the standard (CNN)-based algorithms providing satisfactory outcomes for object detection in general, also very rare approaches used for detecting and recognizing tiny objects such as pests have been detected. CNN-based two-stage recognition and identification approaches for diminutive pests is proposed in this paper. A neighborhood proposal network is presented for YOLOv3 and a re-recognition approach for pest identification is implemented. Also, data augmentation technique by means of image dispensation is used as a last step. Figure 1 shows the architecture of Resnet50. Figure 2 and 3 explains the Training and testing with Resnet50.

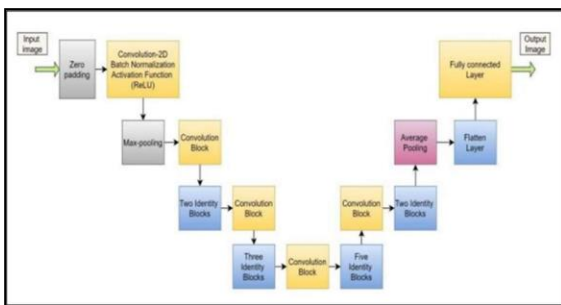


Fig. 1. Architecture of Resnet50

ResNet50 may be a variant of ResNet model which has 48 Convolution layers alongside 1 MaxPool and 1 Average Pool layer. It has 3.8×10^9 Floating points operations. It's a widely used ResNet model [14].

There are different CNN networks have been used for deep learning applications. One among them is Resnet50 architecture [14] which contains zero-padding. Also, it receipts input in 64×64 size. Once 200 epochs are completed the input is in 1×1 scopes. Three convolutions are available in the network which is in the order of 1×1 , 3×3 , 1×1 . The filters or kernel sizes are 64, 128, 256, 512, 1024 and 2048. To yield the high coefficients existing in the input image, max pooling is performed. Convolution and identity blocks (ID) are the first two blocks, trailed by convolution 2D, batch normalisation, and activation functions. The instruction of convolution blocks (CB) and 2, 3, 5 ID is used for iteration. The architecture's output is then obtained using the pooling layer, flatten layer, and dense layers [15-17].

Dark-net is a neural network background written in C and CUDA. It's simple to set up and works with both CPU and GPU processors. Dark-net can handle recurrent neural networks without CUDA or Open-CV, making them effective models for describing data that varies over time [18]. Architecture of yolov3- dark-net network is depicted in figure 4. YOLOv3 object detection dark-net network model is shown in figure 5. The Training and testing with yolov3-dark-net network is shown in figure 6.

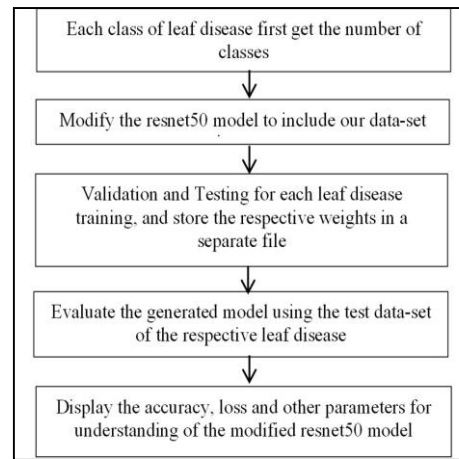


Fig. 2. Training and testing with Resnet50

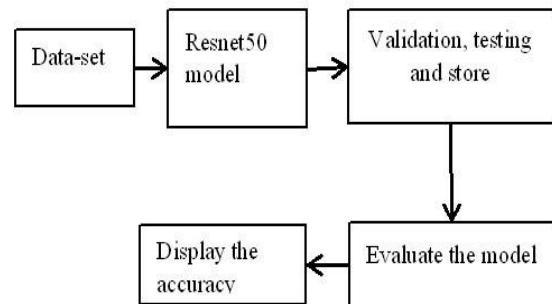


Fig. 3. Convolutional Neural Network- Resnet50

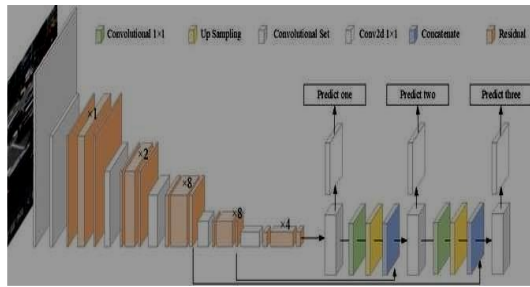


Fig. 4. Architecture of yolov3- dark-net network

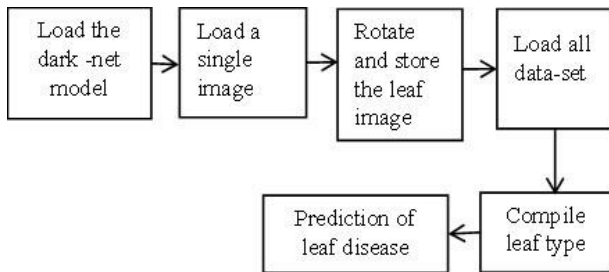


Fig. 5. YOLOv3 object detection dark-net network model

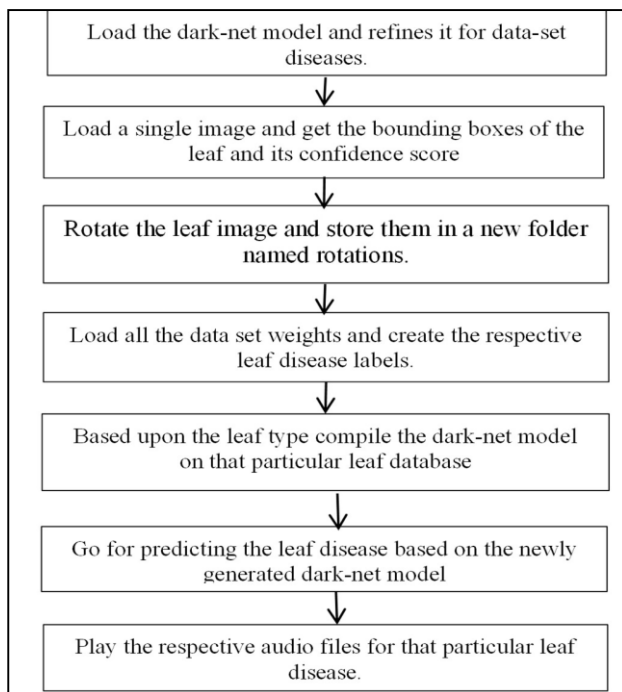


Fig. 6. Training and testing with yolov3- dark-net network

IV. RESULTS AND DISCUSSION

Both Resnet50 and YOLOv3 object detection dark-net network model is implemented using python and executed with set of leaf datasets which are publicly available. The

proposed system is tested with 32 leaf images with and without disease. ResNets work on the premise of building deeper networks compared to other simple networks while concurrently determining an optimal number of layers to avoid the vanishing gradient problem. He et al. proposed

ResNet, which won the 2015 ImageNet competition. Deeper networks can be trained using this strategy [19]. YOLOv3 is You Only Look Once, Version 3 which is an object detection system that detects objects in real time and recognises specific things in films, live feeds, and photos. To detect an item, YOLO uses features learned by a deep convolutional neural network. Joseph Redmon and Ali Farhadi produced the first three versions of YOLO [20]. YOLOv3 object identifier output is shown in Figure 8 and 9. The graph in figure 10 explains the accuracy of YOLOv3 which provides better accuracy than Resnet50.

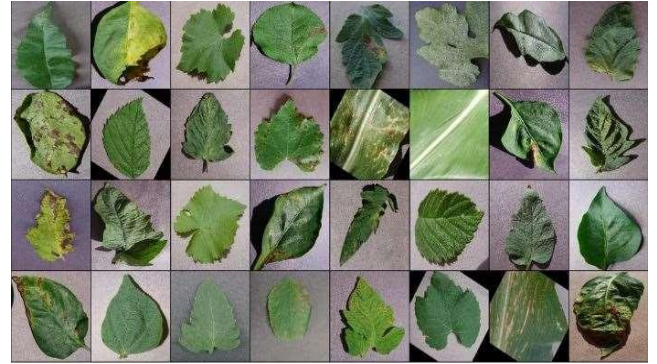


Fig. 7. Sample Dataset

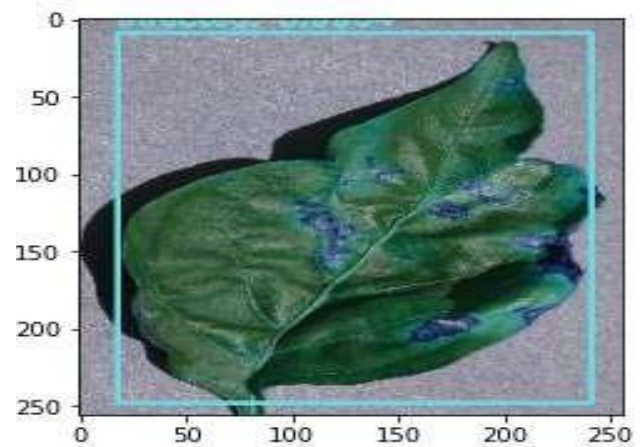


Fig. 8. YOLOv3 object identifier output



Fig. 9. Pepper Leaf Disease Detection

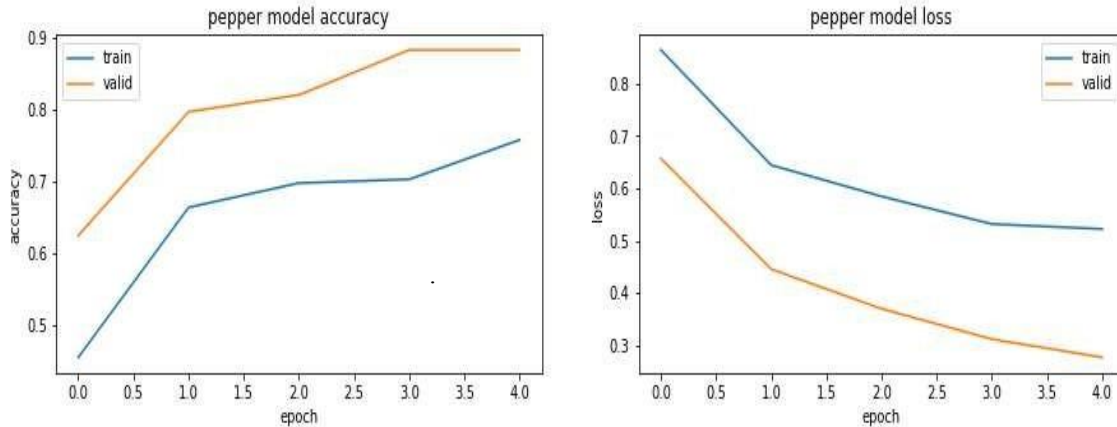


Fig. 10. Accuracy of Pepper Model

V. CONCLUSION

There are several image processing approaches in the literature which are available to pest detection in crops. In this paper plant pests recognition approaches are constructed on deep learning such as Resnet50 and Yolov3. Based on our experiments, Yolov3 provides better accuracy than the Resnet50 network. Plant pest detection technology is very much useful for agricultural applications. Early diagnosis, whether by direct inspection or computer interpretation, is extremely challenging. The importance and desire for early diagnosis is all that is required, which is better for preventing and controlling plant diseases and pests, as well as stopping their spread and growth. When the daylight is adequate, the highest image quality is typically produced, and photographing in overcast weather will raise the complexity of image pre-processing and lower the attractiveness of effect. Furthermore, during the onset of plant diseases and pests, even high-resolution photos are difficult to investigate. To understand the prevalence and forecast of diseases and pests, it is required to combine meteorological and plant protection data such as temperature and humidity.

REFERENCES

- [1] Agriculture and Food, Drones for crop health monitoring, <https://www.maxinai.com/resources/drones-for-crop-health-monitoring-taking-agriculture-to-new-heights>
- [2] Perform Object Detection With YOLOv3, <https://machinelearningmastery.com/how-to-perform-object-detection-with-yolov3-in-keras/>
- [3] Bo Wang, "Identification of Crop Diseases and Insect Pests Based on Deep Learning", *Scientific Programming*, vol. 2022, 2021, <https://doi.org/10.1155/2022/9179998>.
- [4] Mingyuan Xin, Yong Wang, "Image Recognition of Crop Diseases and Insect Pests Based on Deep Learning", *Wireless Communications and Mobile Computing*, vol. 2021, 2021, <https://doi.org/10.1155/2021/5511676>
- [5] M. Kulback, J. Segen, W. Kniec, R. Klempous, K. Kluwak, J. Nikodem, J. Kulbacka, A. Serester, "Survey of Drones for Agriculture Automation from Planting to Harvest", pp. 353-358, 2018, 10.1109/INES.2018.8523943.
- [6] M.G. Selvaraj, A. Vergara, H. Ruiz, N. Safari, S. Elayabalan, W. Ocimati, G. Blomme "AI-powered banana diseases and pest detection", *Plant Methods*, vol. 15, 2019.
- [7] <https://doi.org/10.1186/s13007-019-0475-z>.
- [8] Abraham Chandy, "Pest Infestation Identification in Coconut Trees using Deep Learning", *Journal of Artificial Intelligence and Capsule Networks*, vol.01, no. 01, pp. 10-18, 2019, DOI: <https://doi.org/10.36548/jaicn.2019.1.002>.
- [9] Fernando H Iost Filho, Wieke B Heldens, Zhaodan Kong, Elvira S de Lange, "Drones: Innovative Technology for Use in Precision Pest Management", *Journal of Economic Entomology*, vol. 113, no. 1, February 2020, pp. 1-25, <https://doi.org/10.1093/jee/toz268>.
- [10] D. Gao, Q. Sun, B. Hu, and S. Zhang, "A Framework for Agricultural Pest and Disease Monitoring Based on Internet-of-Things and Unmanned Aerial Vehicles," *Sensors*, vol. 20, no. 5, p. 1487, Mar. 2020, <https://doi.org/10.3390/s20051487>.
- [11] M. Van Hulle, S. Sladojevic, M. Arsenovic, A. Anderla, D. CuD. Stefanovic, "Deep Neural Networks Based Recognition of Plant Diseases by Leaf Image Classification", vol. 22, no. 6, 2016, DOI: 10.1155/2016/3289801.
- [12] Yan Guo, Jin Zhang, Chengxin Yin, Xiaonan Hu, Yu Zou, Zhipeng Xue, and Wei Wang, "Plant Disease Identification Based on Deep Learning Algorithm in Smart Farming", *Discrete Dynamics in Nature and Society*, vol. 2020, 2020, <https://doi.org/10.1155/2020/2479172>.
- [13] S.M. Hassan, A.K. Maji, M. Jasi nski, Z. Leonowicz, E. Jasi nska, "Identification of Plant-Leaf Diseases Using CNN and Transfer-Learning Approach", *Electronics*, vol. 10, 2021, <https://doi.org/10.3390/electronics10121388>.
- [14] X. Tian and C. Chen, "Modulation Pattern Recognition Based on Resnet50 Neural Network," 2nd International Conference on Information Communication and Signal Processing (ICICSP), Weihai, China, pp. 34-38, 2019.
- [15] N. M. Firdaus, D. Chahyati and M. I. Fanany, "Tourist Attractions Classification using ResNet," *International Conference on Advanced Computer Science and Information Systems (ICACSIS)*, Yogyakarta, 2018, pp. 429-433.
- [16] A. S. B. Reddy and D. S. Juliet, "Transfer Learning with ResNet-50 for Malaria Cell-Image Classification," *International Conference on Communication and Signal Processing (ICCSP)*, 2019, pp. 0945-0949, doi: 10.1109/ICCSP.2019.8697909.
- [17] P. Wu and Y. Tan, "Estimation of Economic Indicators using Residual Neural Network ResNet50," *International Conference on Data Mining Workshops (ICDMW)*, Beijing, China, 2019, pp. 206-209.
- [18] S. Inthiyaz, S.H. Ahammad, A. Sai Krishna, V. Bhargavi, D. Govardhan, V. Rajesh, "YOLO (YOU ONLY LOOK ONCE) making object detection work in medical imaging on convolution detection system", *International Journal of Pharmaceutical Research*, vol. 12, no.2, pp. 312-326, 2017.
- [19] ResNet, <https://arxiv.org/pdf/1512.03385.pdf>
- [20] YOLOv3, <https://viso.ai/deep-learning/yolov3-overview/>